

DPP-CQ: Digital Product Passport for Cultural & Quality Goods

Standard Architecture and Core Specifications — Version 2.0.0-draft

International Communication Organization (ICO)

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- DPP-CQ: Digital Product Passport for Cultural & Quality Goods
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DPP-CQ: Digital Product Passport for Cultural & Quality Goods

Standard Architecture and Core Specifications

DPP-CQ 文化与品质数字产品护照

标准架构与核心规范

(ICO Digital Trust Standards Framework v2.0.0-draft)

(ICO 数字信任标准框架 v2.0.0-draft)

Abstract · 摘要

Against the backdrop of deep integration between the digital economy and global trade, cross-border product circulation is evolving from *physical flow* to *trusted flow*. Digital Product Passports (DPPs), as an emerging digital trust infrastructure, have achieved significant progress in the environmental data domain for industrial products, yet notable gaps remain in their coverage of cultural products, specialty agricultural goods, geographical

indication products, and other categories.

This document systematically presents the overall architecture, core specifications, governance mechanisms, and implementation roadmap of the **DPP-CQ Digital Product Passport for Cultural & Quality Goods standards framework**. Built upon mature technologies such as W3C DIDs, Verifiable Credentials (VCs), and standards-based selective disclosure (SD-JWT / BBS+), this framework focuses on establishing digital trust in the cultural value and quality dimensions, addressing the inclusivity gap in existing DPP standards ecosystems.

The core standard, **ICO Std 2001 (DPP-CQ)**, is a digital passport specification for cultural and quality products. It supports dual carriers of QR codes and NFC, provides a three-tier verification mechanism, and protects sensitive craftsmanship and commercial data through selective-disclosure proofs (SD-JWT / BBS+). The standards framework employs a multi-stakeholder governance model to ensure openness, neutrality, and inclusivity.

This document is issued as a Public Review Draft. We sincerely invite governments, enterprises, academic institutions, and non-governmental organizations worldwide to participate in building a more inclusive global digital trust ecosystem.

v2.0.0-draft update (Aug 2026): Following the first public review, this revision substantially expands the standard's international interoperability and compliance posture: a GS1 Digital Link / GTIN interop profile, three-tier physical carrier specification (open QR + SDM NFC + tamper-evident NFC) aligned with EN 18220, two normative credential format profiles (W3C Data Integrity with BBS+ and IETF SD-JWT VC / RFC 9529), an optional Chinese national cryptography suite (SM2/SM3/SM4), an ISO 14067-aligned sustainability data module (quality + sustainability dual-dimension model), AI-assisted assessment transparency provisions aligned with GB/T 47507-2026 and EU AI Act Art. 50, data lifecycle governance metadata, and UNTP-style conformity claims. All v2.0 additions are backward compatible with v1.x credentials.

Keywords: Digital Product Passport; Cross-Cultural Digital Trust; Quality Assets; DPP-CQ; Decentralized Identifiers; Verifiable Credentials; Selective Disclosure (SD-JWT / BBS+); GS1 Digital Link; Multi-Stakeholder Governance

Scope · 范围

This document describes the **overall architecture, core specifications, governance mechanisms, and implementation roadmap** of the **DPP-CQ Digital Product Passport for Cultural & Quality Goods standards framework**, with particular focus on the technical framework and application model of the core product-level standard **ICO Std 2001-2026 (DPP-CQ)**.

Applicability: - Categories with cultural value and quality attributes, including cultural products, Intangible Cultural Heritage-related products, Geographical Indication (GI) products, specialty agricultural goods, and haute couture products - Application scenarios including product identity verification, quality grading, traceability verification, and digital confirmation of cultural value - Technical integration and system interoperability for all participating parties, including producers, certification bodies, trading platforms, and consumers

Exclusions: - Highly regulated special categories such as pharmaceuticals and medical devices (which must comply with sector-specific regulatory requirements) - High-sensitivity scenarios at the financial-grade or personal safety level, such as financial payment and identity authentication - Replacement of mandatory compliance requirements such as statutory product safety and quality inspection in respective jurisdictions

Purpose · 目的

This document aims to: 1. Systematically present the design philosophy and overall architecture of the ICO standards framework, providing a panoramic understanding for all stakeholders; 2. Define the core technical

specifications and interoperability requirements of the DPP-CQ standard, guiding technical implementation; 3. Describe the governance mechanisms and participation pathways of the standards framework, inviting global stakeholders to co-build; 4. Solicit feedback from all sectors of society to promote continuous improvement and widespread adoption of the standards.

Status of This Document · 本文档状态

This document is a **Public Review Draft** published by the International Communication Organization (ICO).

The content of this document may be revised based on community feedback. The official version shall be as officially published by ICO. This v2.0.0-draft incorporates the results of the first public review (July 6 – August 20, 2026) and opens a **second review period: September 1 – October 15, 2026**, focused on the new international interoperability and compliance annexes. Feedback may be submitted through the following channels:

- GitHub Issues: <https://github.com/ICO-cloud/dpp-cq-standard/issues>
- Email: info@icoun.org

This document does not constitute a legally binding instrument, nor does it represent the official position of ICO members. Formal publication of the standard requires approval by the ICO Standards Council.

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Reference implementations and software source code are released under the **Apache License 2.0** open-source license.

Editors and Contributors · 编辑与贡献者

Technical Editor: ICO Technical Secretariat **Standards Working Groups**: DPP-CQ Technical Working Group, Governance & Compliance Model Group, Interoperability Working Group

Contributing Organizations (in alphabetical order): - International Communication Organization (ICO) - 天机可信价值研究院 (Tianji Credibility Research Institute) - (Additional organizations to be added)

For the complete list of contributors and detailed revision history, please refer to the commit history of the GitHub repository.

Revision History · 修订历史

Version	Date	Description of Changes
v2.0.0	-	International interoperability & compliance expansion: GS1 Digital Link/GTIN interop profile; three-tier carrier specification (EN 18220 aligned); SD-JWT VC (RFC 9529) profile alongside BBS+; optional

v2.0.0-draft	Aug 2026	SM2/SM3/SM4 national cryptography suite; ISO 14067 sustainability data module (quality+sustainability dual-dimension); AI-assisted assessment transparency (GB/T 47507-2026, EU AI Act Art. 50); data lifecycle governance metadata; UNTP-style conformity claims; conformity declarations and PIA summary published; all additions backward compatible with v1.x
v1.3	July 2026	Format standardization: aligned with international technical specification formats; added abstract, document status, copyright notice, and revision history; annexes classified as normative/informative; unified section numbering; main title updated to DPP-CQ; verified accuracy of data citations
v1.2	July 2026	Added compliance and legal framework chapter, DPP-CQ technical specification annex, intellectual property policy, standards version management strategy, and FAQ; strengthened references to UNESCO/WIPO/UNECE
v1.1	July 2026	Framework restructuring and refinement: optimized architecture description; added risks and challenges chapter, pilot framework, and UNECE interoperability initiative references
v1.0	July 2026	First public release (Public Review Draft)
v0.9	July 2026	Internal working draft

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1. Introduction · 前言

In an era of deep integration between the digital economy and global trade, cross-border product circulation is transitioning from *physical flow* to *trusted flow*. However, significant **inclusivity gaps** exist in the current global digital product verification ecosystem:

- Existing Digital Product Passport (DPP) standards primarily focus on environmental and circular economy data for industrial products, and do not adequately cover categories such as cultural products, specialty agricultural goods, and geographical indication products.
- A vast number of products with unique cultural value and quality advantages lack digital trust-transmission tools, making it difficult for them to fully realize their value in global markets. According to a joint

OECD/EUIPO study, global trade in counterfeit goods reached USD 467 billion in 2021, accounting for 2.3% of global imports. Among these, geographical indication products and specialty cultural products, with their high recognizability and premium pricing potential, have become prime targets for counterfeiting. Specialty products from developing countries are severely undervalued due to lack of trust.

- Developing economies face multiple challenges in digital trust infrastructure development, including technological, standardization, and capacity gaps. The UN *Global Digital Compact* (GDC) explicitly calls for increased support for digital capacity building in developing countries to ensure that digital technologies benefit all people.

The International Communication Organization (ICO), as a multi-stakeholder international cooperation platform, is committed to advancing **Global Digital Inclusion** and **Cross-Cultural Digital Trust**. This document presents the overall architecture, core standards, governance mechanisms, and implementation roadmap of the ICO standards framework, and invites global stakeholders to join us in building a more inclusive global digital trust ecosystem.

2. Mission & Vision · 使命与愿景

2.1 Mission

To empower the digital preservation and value transmission of global cultural and quality assets through open standards.

以开放标准赋能全球文化与品质资产的数字化保护与价值传递。

2.2 Vision

To build a **globally applicable, technology-neutral, and culturally inclusive** digital trust standards framework, ensuring that every product carrying cultural value and quality commitment possesses a verifiable, traceable, and mutually recognizable digital identity. By 2030, achieve 100,000+ cultural and quality products with globally mutually recognizable digital identities, driving global trade from *price competition* to *value competition*.

2.3 Core Principles

Principle	Description	Practical Manifestation
Open	Standards are open to all stakeholders, regardless of country, region, or ownership structure	Free adoption with no membership fee barriers
Neutral	Technology-agnostic architecture, not tied to any specific vendor or technology stack	Supports multiple blockchain and cloud platforms
Inclusive	Fully considers the needs of different cultural backgrounds and development stages	Supports lightweight implementations for low-resource environments
Interoperable	Based on international technical standards, ensuring connectivity with existing and future systems	Compatible with W3C, ISO, GS1, UNECE, and other standards
Trustworthy	Verifiable data, traceable processes, and adjudicable disputes	Three-tier verification + Root Resolver Network + selective-disclosure privacy protection
Privacy by Design	Privacy protection embedded in architectural design; data minimization	Selective disclosure (SD-JWT / BBS+)

3. Background & Rationale · 背景与必要性

3.1 The Trust Deficit in Global Trade

In current cross-border global trade, categories such as specialty agricultural goods, cultural products, and handicrafts face pervasive trust challenges: consumers struggle to verify the true origin, quality grade, and cultural background of products; purchasers lack efficient and reliable supply chain traceability tools; authentication of geographical indication products is costly; and cultural value and craft heritage cannot be effectively transmitted through traditional channels.

A direct consequence of this trust deficit is the “bad money drives out good money” effect — genuinely superior products and producers who uphold traditional craftsmanship are unable to obtain returns commensurate with their value in the global market.

3.2 Coverage Gaps in Existing Standards

Existing DPP standards are primarily led by advanced economies, focusing on environmental footprints and circular economy data for industrial products, compliance-driven mandatory information disclosure, and serving the market access requirements of developed economies. Significant standard gaps remain in the following areas:

Gap Area	Current State	ICO Positioning
Digital confirmation of cultural product rights	No unified cultural value data standard	Establish cultural data models and verification specifications
Cross-border mutual recognition of Geographical Indications	WIPO has legal framework but no technical standards	Develop GI digital mutual recognition technical specifications
Quality grading for specialty agricultural goods	Individual national standards, no global unified digital framework	Establish cross-cultural quality assessment methodology
Digital capacity in developing economies	High technical barriers, high costs, low participation	Provide lightweight, low-threshold solutions

3.3 Interoperability with Existing Standards

ICO standards adopt a strategy of “**baseline compatibility, differentiated value**”:

- **W3C DID/VC**: Used as the underlying identity and credential technology foundation, fully compatible; v2.0 supports both Data Integrity proofs (BBS+) and SD-JWT VC (RFC 9529)
- **GS1 Identification System**: GTIN/GLN identifiers and GS1 Digital Link resolution are supported through a normative interop profile; supply-chain events map to EPCIS 2.0/CBV 2.0; DID and GTIN are bound as complementary identifiers
- **EU DPP / ESPR / EN 18219-18223**: Open data carriers (EN 18220), unique identifiers (EN 18219) and sustainability data attributes are aligned at field level; conformity claims use UNTP-style vocabulary
- **UN/CEFACT UNTP**: Conformity claim and interoperability vocabulary adopted as the multilateral bridge for cross-regime data exchange
- **ISO standards**: Compatible with ISO 22000 (food safety), ISO 14067 (product carbon footprint), ISO 26000 (social responsibility), ISO 22739 (blockchain terminology)
- **Chinese national standards**: AI trustworthiness (GB/T 47507-2026) and commercial cryptography (SM2/SM3/SM4) supported for domestic deployments

DPP-CQ does not replace statutory conformity assessment or sectoral certification held by producers; it carries and references such attestations in a verifiable, machine-readable form.

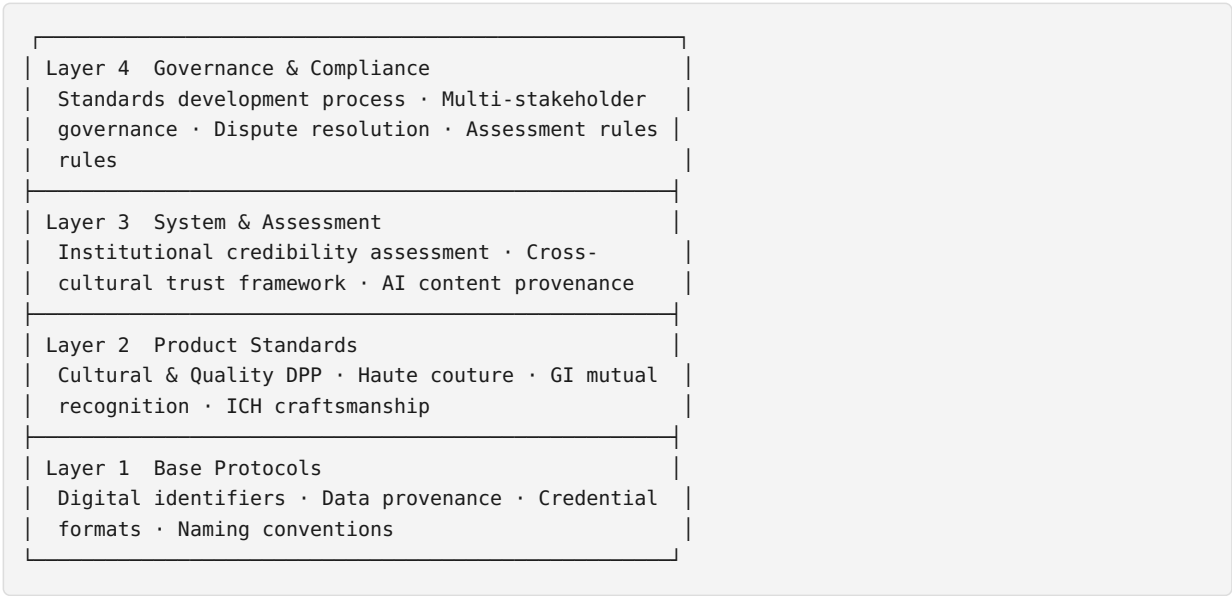
3.4 Technology Maturity Window

Technologies such as W3C DIDs (Decentralized Identifiers), Verifiable Credentials, blockchain-based evidence anchoring, and selective-disclosure credential technology (SD-JWT, BBS+) have entered mature application phases, providing the technical foundation for building a globally applicable open standard for the cultural and quality dimensions.

Meanwhile, international documents and data such as UNESCO's *Operational Directives for the Implementation of the Convention for the Safeguarding of the Intangible Cultural Heritage* (2016/2018 edition) and WIPO's *World Intellectual Property Indicators 2024* all indicate the urgency and necessity of leveraging digital technology to protect cultural assets and geographical indications. The timing is opportune for establishing open standards oriented toward the cultural and quality dimensions.

4. Standards Architecture · 标准体系架构

4.1 Four-Layer Architecture



4.2 Standards Numbering System

Series	Layer	Positioning	Number Range
1000 Series	Base Protocols Layer	Underlying technical specifications, common foundation for all upper-layer standards	1001–1099
2000 Series	Product Standards Layer	Specific data standards and verification specifications for different product categories	2001–2099
3000 Series	System & Assessment Layer	Standards at the institutional assessment, trust system, and methodology level	3001–3099
4000 Series	Governance & Compliance Layer	Standards development processes, governance rules, dispute resolution mechanisms	4001–4099

4.3 Initial Core Standards Inventory

Standard Number	Standard Name	Status	Expected Publication
ICO Std 1001-2026	Global Information Integrity & Data Provenance	Draft	2026 Q3
ICO Std 1002-2026	Digital Credential Format Specification	Under Development	2026 Q3
ICO Std 2001-2026	Digital Product Passport for Cultural & Quality Goods (DPP-CQ)	Public Review Draft	2026 Q3
ICO Std 2002-2026	Global Haute Couture & Cultural Attire Quality Standard	Planned	2026 Q4
ICO Std 2003-2026	Global Framework for GI & Origin Digital Mutual Recognition	Planned	2027 Q1
ICO Std 3001-2026	Global Digital Credibility Assessment (GDCC)	Draft	2026 Q4
ICO Std 4001-2026	Standards Development & Publication Process	Under Development	2026 Q3
ICO Std 4002-2026	Multi-Stakeholder Governance Charter	Under Development	2026 Q3

5. Core Standards · 核心标准介绍

5.1 ICO Std 2001: Digital Product Passport for Cultural & Quality Goods (DPP-CQ)

Positioning: A digital identity standard for cultural products, specialty agricultural goods, geographical indication products, and ICH handicrafts, providing a unique, trustworthy, and verifiable digital identity credential for every product carrying cultural value and quality commitment.

Core Data Dimensions:

Data Module	Core Fields	Required/Optional
Base Identity	Product unique identifier (DID), issuer information, validity period, category	Required
Quality Data	Quality grade, test report summary, certification credentials, batch information	Category-required
Cultural Data	Cultural background, craft heritage, ICH level, artisan/transmitter information	Category-required
Geographical Indication	Origin, GI certification number, production region data	Required for GI products
Traceability Data	Full lifecycle event records (production/processing/testing/distribution)	Optional
Credential Data	Verifiable credential metadata, signatures, revocation status	Required

Technical Features:

- Based on W3C DID Core and Verifiable Credentials international standards
- Supports dual-carrier verification via QR code and NFC
- Three-tier verification mechanism: Quick Verification (<1s) / Deep Verification / Judicial-Grade Verification
- Hybrid storage architecture: data stored off-chain, hashes anchored on-chain
- **ZKP-enabled Privacy:** The standard enables manufacturers, ICH transmitters, and distribution entities to prove compliance with specific quality grades and origin regulatory requirements to consumers and regulatory authorities through selective-disclosure proofs (SD-JWT / BBS+), without disclosing core craft recipes, supply chain node identities, or commercial procurement prices — achieving “data usability without visibility, trustworthiness without information leakage”
- Selective Disclosure: Data holders may choose the scope of disclosure based on the verification scenario

Applicable Categories: Geographical indication agricultural products (tea, grains & oils, fruits, medicinal herbs, spices), Intangible Cultural Heritage products (ceramics, embroidery, lacquerware, woodwork), haute couture and cultural attire, premium food and beverages, handicrafts and cultural creative products.

5.2 ICO Std 1001: Global Information Integrity & Data Provenance

Defines the credibility assessment framework and provenance methodology for digital content and data assets, with particular emphasis on provenance identification specifications for AI-generated content. Covers data integrity verification, provenance chain construction methods, credibility grading assessment, timestamps, and evidence anchoring interfaces.

5.3 ICO Std 3001: Global Digital Credibility Assessment (GDCC)

A digital credibility assessment and evaluation standard for institutions (enterprises, media, NGOs, etc.), constructing an assessment system across multiple dimensions including information transparency, data credibility, governance compliance, and cultural responsibility, with support for cross-cultural scenario adaptation.

6. Governance · 治理架构

6.1 Multi-Stakeholder Governance Model

The ICO standards framework employs a multi-stakeholder governance model to ensure the openness, representativeness, and credibility of standards. Important decisions follow a consensus-first principle; where consensus cannot be reached, decisions are adopted by a two-thirds majority.

Governance design references best practices from international standards bodies such as W3C and IETF, ensuring: - Balanced representation of economies at different development stages - Diverse participation from government, industry, academia, and civil society - Effective separation of technical decision-making from commercial interests

6.2 Governance Structure

General Assembly (supreme decision-making body): Convened annually to deliberate on strategic direction, budget, and Council elections.

Standards Council (decision-making body): Responsible for strategic decisions, standard approval, budget approval, and final dispute adjudication. Seats are allocated by category: Government Representatives 25% / Industry 25% / Academia 20% / Civil Society 20% / Technical Secretariat 10% (non-voting).

Specialized Committees: - Technical Committee: Responsible for standard technical review and technical roadmap planning - Strategy Committee: Responsible for strategic planning and external cooperation -

Compliance & Dispute Committee: Responsible for compliance review and dispute adjudication

Standards Working Groups: - WG-DPP: Digital Product Passport Working Group - WG-GI: Geographical Indication Mutual Recognition Working Group - WG-DID: Decentralized Identifiers and Credentials Working Group - WG-GDCC: Institutional Credibility Assessment Working Group - WG-Governance: Governance Rules Working Group

6.3 Technical Secretariat

The Technical Secretariat is the neutral executive body of the standards framework, undertaken by a designated entity authorized by the Standards Council. Its core responsibilities: - Maintain reference implementations and open-source code repositories for the standards - Provide technical support and conformance testing platforms - Operate the root resolution infrastructure and trusted registries - Support the technical work of working groups - Organize technical training and developer community building

Key Principle: The Technical Secretariat does not participate in standard voting, ensuring the separation of technical execution from standards development.

6.4 Infrastructure and Root Resolution Mechanism

To ensure high security and continuous availability of data resolution in global cross-border circulation, the ICO Council authorizes the Technical Secretariat to build and operate:

- **ICO Root Resolver Network:** A globally distributed DID resolution infrastructure providing highly available, low-latency identifier resolution and credential status query services
- **Tianji Credibility Registry (TCR):** A unified credential hash registration and anchoring database

Operating Mechanism: When any third-party certification body or service provider issues Digital Product Passports under ICO Std 2001/2003, the credential hash fingerprints and identifiers (DIDs) generated must be registered and consensus-anchored at the root nodes, ensuring traceability and judicial-grade authenticity verification across networks and geographical regions worldwide.

Security Assurance: - Multi-region redundant deployment with 99.99% availability target - At least one independent third-party security audit per year, with audit reports made public - Major technical upgrades require review by the Technical Committee and approval by the Standards Council

7. Technical Approach · 技术路线

7.1 Technology Selection Principles

- **International Standards First:** Prioritize adoption of published international standards (W3C, IETF, ISO)
- **Open Source & Open:** Core technology stack based on open-source licenses
- **Extensible Architecture:** Modular design supporting future standard extensions
- **Privacy by Design:** Data minimization principle, implemented via standards-based selective disclosure (SD-JWT / BBS+)

7.2 Core Technology Stack

Layer	Technical Standard	Reference / Compatible Specification
Decentralized Identifiers	W3C DID Core 1.0	W3C Recommendation (Jul 2022)
Verifiable Credentials	W3C Verifiable Credentials Data Model v2.0	W3C Recommendation
		W3C Recommendation (Jul 2022)

Selective Disclosure	SD-JWT (RFC 9529) / BBS+ signatures	IETF RFC 9529; W3C Data Integrity BBS cryptosuite
Data Semantics	JSON-LD 1.1	W3C Recommendation / IETF BCP 19
Cryptographic Signatures	BBS+ / Ed25519 / ECDSA P-256; SM2+SM3 optional	IETF RFC 8032; GB/T 32918 / GB/T 32905 (national crypto option)
Global Product Identification	GS1 GTIN / GLN / GS1 Digital Link 1.6	GS1 General Specifications; EU DPP EN 18219 alignment
Supply Chain Events	GS1 EPCIS 2.0 / CBV 2.0	GS1 Standard (2022); mapping profile defined in v2.0
Data Anchoring	Cryptographic Hash Anchoring	Blockchain-agnostic, multi-chain support
Physical Verification	QR ISO/IEC 18004 (open carrier); NFC ISO/IEC 14443 with SDM (NTAG 424 DNA class)	EN 18220:2026 alignment; NFC Forum NDEF
Sustainability Data	Product carbon footprint per ISO 14067	Optional sustainability module; ESPR data-attribute direction
Interface Specification	RESTful API / OpenAPI 3.0; Digital Link content negotiation	OpenAPI Initiative; GS1 Digital Link resolver pattern

7.3 Multi-Chain and Multi-Cloud Architecture

ICO standards are not bound to any specific blockchain or cloud platform. Through a unified hash anchoring abstraction layer, multiple underlying storage solutions are supported (public chains, consortium chains, distributed hash tables, trusted cloud storage, etc.). Different jurisdictions may select anchoring solutions compliant with local regulations while maintaining cross-chain verification consistency.

7.4 Open Source Strategy

Component	Open Source License	Description
Standards documentation	CC BY 4.0	Free to use, reproduce, and adapt; attribution required
Base protocol reference implementations	Apache 2.0	Fully open, business-friendly
SDKs and development tools	Apache 2.0	Multi-language SDKs, encouraging ecosystem participation
Contribution mechanism	CLA	Contributor License Agreement ensuring clear intellectual property rights

7.5 Intellectual Property Policy

The ICO standards framework adopts an intellectual property policy of “openness first, reasonable protection”:

Standard-Essential Patent (SEP) Policy: - Encourages patent holders to license their essential patents on FRAND (Fair, Reasonable, and Non-Discriminatory) terms - Core standards (DIDs, credential formats, and other base layers) prioritize open-source technology solutions free of patent encumbrances - Patent searches and disclosures are conducted prior to standard publication to minimize potential patent risks - An Intellectual Property Working Group is established to oversee patent policy development and dispute resolution

Trademark and Logo Policy: - “ICO Verified” mark, “DPP-CQ” mark, and others are registered trademarks of ICO - Products and services meeting DPP-CQ conformance requirements may use official marks subject to compliance

requirements; DPP-CQ conformance is an assessment/verification (评定/验证), not statutory certification (认证) - Logo usage specifications and supervision mechanisms are established to prevent abuse and misleading - Logo usage by regional nodes and partners requires authorization and review

Copyright Policy: - Copyright in standard texts is held by ICO and made available under the CC BY 4.0 license - Copyright in reference implementations and tool code is held by contributors and open-sourced under the Apache 2.0 license - Third-party derivative works must comply with the corresponding open-source licenses and indicate source

7.6 Standards Version Management

Version Numbering Scheme: Uses the format “major.minor.patch” (e.g., v1.2.3) - **Major version:** Architectural expansion of the standard’s scope (e.g. v2.0 adds the GS1 interop profile, SD-JWT VC format and sustainability module); all v2.0 data additions are OPTIONAL so v1.x credentials remain valid - **Minor version:** Backward-compatible functional additions within the same architecture - **Patch version:** Backward-compatible bug fixes

Backward Compatibility Commitments: - Minor version updates maintain backward compatibility of data formats and interfaces - Major version updates provide at least 12 months of transition period and migration tools - The validity of issued credentials is not affected by standard version updates

Standards Lifecycle: - Working Draft (WD) → Committee Draft (CD) → Public Review Draft (PR) → International Standard (IS) - International standards are reviewed every 5 years to ensure continued validity - Obsolete standards initiate revocation procedures with advance notice to all adopters

7.7 Physical Carrier Strategy: Three-Tier, Open by Default

DPP-CQ v2.0 defines three physical carrier levels (normative detail: [Carrier Specification](#)):

Level	Carrier	Security profile	Reader requirement
L1	Open QR (GS1 Digital Link in interop profile)	Cloneable; authenticity via registry	Any smartphone camera
L2	SDM secure NFC (NTAG 424 DNA class)	Per-scan cryptographic SUN authentication	Any NFC phone, NDEF open read
L3	Tamper-evident NFC (TagTamper class)	Tamper detection + SDM	Any NFC phone

Two principles govern carrier design:

1. **Open carrier principle (aligned with ESPR Art. 9 and EN 18220):** every product must be readable with generally available means — no proprietary app, registration, or fee. Security is achieved through cryptographic verification of the resolved data, never through hiding the payload.
2. **Dual carrier for international products:** products in the GS1 interop profile carry both an open QR (universal scanning, retail/customer flows) and an SDM NFC tag (cryptographic authenticity); both resolve to the same passport, and the verification page indicates which carrier was presented.

7.8 Dual-Dimension Data Model: Quality + Sustainability

Existing DPP regimes (notably the EU ESPR framework) organize product data primarily around environmental sustainability and circular economy. DPP-CQ’s core differentiation is the **quality and cultural value** dimension. v2.0 formalizes a dual-dimension model rather than choosing between them:

- **Quality & cultural dimension (DPP-CQ core):** quality grade, sensory and laboratory attributes, craftsmanship, ICH/GI status, artisan lineage, assessment methodology and AI-use disclosure.
- **Sustainability dimension (v2.0 optional module):** product carbon footprint per ISO 14067, material composition with recycled content, end-of-life guidance — structured to map onto ESPR delegated-act data

attributes as they are adopted per product class.

This design lets a single credential serve both purposes: it satisfies the cultural/quality trust gap DPP-CQ was created to address, while producing machine-readable sustainability data that EU importers, retailers and regulators can consume. The quality assessment itself remains governed by ICO standards (notably Std 3001), preserving the framework’s independence.

7.9 AI-Assisted Assessment Transparency

Where AI systems support quality assessment (data extraction, pre-scoring, image recognition), DPP-CQ adopts transparency and human-oversight rules aligned with **GB/T 47507-2026** (AI trustworthiness general rules) and the transparency direction of **EU AI Act Art. 50**:

- AI participation **MUST** be disclosed in the credential (`qualityAttributes.assessment.aiDisclosure`).
- The final grading decision **MUST** be made by a qualified human assessor; the credential identifies the decision role and review process.
- The assessment methodology reference and full report **MUST** be retrievable via the credential (`reportReference`).

8. Compliance & Legal Framework · 合规与法律框架

8.1 Data Sovereignty and Cross-Border Compliance Architecture

ICO standards employ an architectural design of “**data localization, hash cross-border**”, fundamentally reducing cross-border data compliance risks:

- **Raw data stored locally**: Detailed product quality, craftsmanship, and supply chain data are stored locally by the issuer or certification body, with no mandatory cross-border transmission
- **Hash fingerprint anchoring**: Only hash fingerprints of data (one-way irreversible) are registered and anchored at root nodes; hashes themselves contain no identifiable original information
- **Selective-disclosure verification**: Sensitive data is verified via SD-JWT / BBS+ selective disclosure; verifiers receive only the claims or compliance conclusions they are authorized for, without access to undisclosed raw data

This architecture is compatible with China’s *Data Security Law* and *Personal Information Protection Law*, the EU’s General Data Protection Regulation (GDPR), and the data sovereignty principles of most countries.

8.2 Compliance Adaptation by Major Jurisdictions

Jurisdiction	Core Compliance Requirements	ICO Standard Adaptation
China	Data classification and grading, security assessment for outbound transfer of important data, critical information infrastructure protection; commercial cryptography requirements; AI trustworthiness (GB/T 47507-2026)	Localized data storage, hash cross-border, independent deployment of domestic root nodes; optional SM2/SM3/SM4 cryptography suite; AI assessment transparency & human-oversight provisions; data lifecycle governance metadata
European Union	GDPR data protection; EUPR digital product passport & sustainability data; EN 18219/18220 series; EU AI Act transparency	SD-JWT/BBS+ selective disclosure, data minimization; GS1 Digital Link & EN 18220-aligned open carriers; ISO 14067 sustainability module; conformity claims; AI-use disclosure
Southeast Asia	Significant variation in national data protection laws, differing localization	Flexible regional node deployment, configurable data storage strategies;

Southeast Asia	protection laws, differing localization requirements	lightweight L1 QR carrier for cost sensitivity
Middle East / GCC	Developing data localization and halal/cultural-product traceability regimes; strong interest in digital trade infrastructure	Regional node deployment; cultural-quality framing compatible with heritage-product protection; GS1/UNTP interop for trade
Global South	Relatively weak data regulations, insufficient digital capacity	Lightweight solutions, capacity building support, technology transfer; zero-licensing-fee adoption (see three red lines: no charging for standards use)

8.2b International Interoperability Position

DPP-CQ is an **independent international standard** and does not subordinate itself to any regional regulatory regime. Its interoperability strategy is “baseline compatibility, differentiated value”:

- **Baseline compatibility (non-negotiable interop):** open data carriers, global identifiers (GTIN alongside DID), standard credential formats (W3C VC / SD-JWT VC), standard APIs, and machine-readable conformity claims. These ensure DPP-CQ credentials can be read and cross-checked by EU DPP, retail, customs and GS1-based systems.
- **Differentiated value (kept independent):** the quality-and-culture assessment methodology (Std 3001), the multi-stakeholder governance model, affordability for small producers, and applicability to cultural product categories that environmental DPPs do not cover.
- **Multilateral bridge:** active engagement with the UN/CEFACT UNTP (UN Transparency Protocol) conformity-credential vocabulary and the ISO DPP work item (ISO/PWI 25534-1 track), so that DPP-CQ interops through neutral, multilateral standards rather than through any single jurisdiction’s registry.

DPP-CQ’s flagship categories (tea, wine & spirits, cultural crafts, traditional medicine and health products, specialty agri-foods) are not within the first ESPR priority product groups; the standard therefore has a strategic window to mature its interop profile before any statutory DPP obligations potentially reach its categories.

8.3 Data Lifecycle Governance

Every DPP-CQ credential in v2.0 carries a `dataLifecycle` record identifying the data controller, retention policy, storage jurisdiction, cross-border mode (default: hash-only), deletion/anonymization rules, privacy notice URI, and PIA/DPIA summary URI. A public PIA summary is maintained at [docs/compliance/pia-summary.md](https://docs.compliance/pia-summary.md), and conformity declarations against GB/T 47507-2026, the ESPR/EN framework, and data-protection regimes are maintained at [docs/compliance/conformity-declarations.md](https://docs.compliance/conformity-declarations.md).

8.4 Electronic Evidence and Legal Effect

ICO standards are themselves technical and industry standards, without legal force (consistent with the nature of international standards such as those from ISO). However, their design fully considers judicial evidence requirements:

- **Hash integrity proof:** Cryptography-based data integrity verification can serve as electronic evidence in most jurisdictions
- **Trusted timestamps:** Integration with authoritative timestamp services meets the temporal certainty requirements for electronic evidence
- **Complete evidence chain:** Issuance chains, change records, and revocation logs constitute a complete evidence chain
- **Third-party audits:** Root nodes and core services undergo regular independent third-party audits to enhance evidentiary credibility

Specific legal effect varies by jurisdiction. ICO encourages regional nodes to collaborate with local legal

institutions to promote recognition of the evidentiary weight of DPP-CQ credentials in their respective jurisdictions.

8.5 Dispute Resolution Mechanism

Referencing UNCITRAL (United Nations Commission on International Trade Law) arbitration rules and international commercial arbitration practices, ICO establishes a three-tier dispute resolution mechanism:

1. **Self-Help Mediation:** Parties negotiate independently through platform tools
2. **Expert Review:** A review panel composed of industry and technical experts issues a ruling
3. **Arbitration (Final):** Submission to international commercial arbitration, with arbitral awards enforceable globally under the New York Convention

9. Getting Involved · 参与方式

9.1 Membership Categories

Membership Category	Core Rights	Primary Obligations	Eligible Applicants
Founding Member	Council seats, priority participation, standard proposal rights	Annual contribution commitment, governance participation	Core founding institutions
Contributing Member	Working group participation, standard proposal rights, voting rights	Abide by charter, contribute to standard content	Enterprises, NGOs, academic institutions
Adopting Member	Free standard use, technical support, listing in ecosystem directory	Logo usage compliance, feedback on usage	Product manufacturers, service providers
Community Member	Participate in open-source community, submit feedback, attend public meetings	No mandatory obligations	Developers, researchers, individuals

9.2 Participation Pathways

Technical Participation: Join the GitHub open-source community and contribute code, documentation, and technical solutions

Standards Participation: Apply to join relevant standards Working Groups (WGs) to participate in standard development and review

Ecosystem Participation: Become a verified service provider, regional node operator, or joint laboratory partner

Adoption Participation: Implement ICO standards in products or services and apply for DPP-CQ conformance assessment

10. Roadmap & Pilots · 路线图与试点规划

10.1 Three-Year Development Roadmap

2026: Foundation and Ecosystem Launch Year - Q3: Publish governance standards and DPP-CQ v1.0 - Q3: Establish Technical Secretariat, launch Root Resolver Network - Q4: Host Global Digital Credibility Governance

Summit - Q4: Publish GDCC standard draft - Year-end target: 3+ standards published, 50+ member institutions, 1,000+ pilot products

2027: Ecosystem Expansion and Regional Deployment Year - Publish 2–3 new standards (GI mutual recognition, haute couture, etc.) - Launch 3 regional nodes: Southeast Asia, Middle East, Africa - Establish 5+ industry joint laboratories - 500+ institutional adopters, 10,000+ verified products - Establish formal cooperation with 2–3 international standards organizations

2028 and Beyond: Global Mutual Recognition and Ecosystem Maturity Year - 10+ standards published - 20+ national/regional nodes - 1,000+ institutional adopters, 100,000+ verified products - Establish mutual recognition mechanisms with major international standards organizations

10.2 Initial Pilot Framework

The first batch of flagship product assessment pilots will launch in the second half of 2026:

Pilot Category	Flagship Direction	Collaboration Model	Success Metrics
GI Tea	Top-tier production region GI tea	Production region + brand joint assessment	Number of verified SKUs, traceability data coverage rate
ICH Ceramics	Handcrafted technique heritage	Artisan + production region joint	Cultural data completeness, user verification volume
Haute Couture Attire	Chinese haute couture brands	Brand assessment + craft evidence anchoring	Full lifecycle passport coverage, market feedback
ICH Handicrafts	Embroidery / lacquerware, etc.	Transmitters + institutional joint	ICH value display, cross- border transaction improvement

Pilot evaluation dimensions: number of verified products, traceability data quality, user verification frequency, cross-border transaction improvement, producer value gain.

11. Risks & Challenges · 风险与挑战

Risk	Description	Mitigation Strategy
Adoption Resistance	SMEs have limited digital capabilities and concerns about integration costs	Provide lightweight toolkits, SaaS platforms, capacity building programs; free during the first 6-month pilot period
Privacy and Data Security	High protection requirements for cultural craftsmanship and commercial data	Strict selective-disclosure application, off-chain storage architecture, regular third-party security audits; data control rests with issuers
Standards Fragmentation	Multiple sets of DPP standards coexist globally, creating confusion for enterprises	Actively promote mutual recognition and compatibility; participate in multilateral interoperability initiatives such as UNECE; avoid closed systems
Governance Neutrality	International community has concerns about the fairness of standard-setting	Transparent governance processes, multi-party oversight mechanisms, independent third-party audits, balanced representation
Technology Evolution Risk	Cryptography and blockchain technologies evolve rapidly, risking	Modular design, extensible architecture, periodic technical reviews,

	standard obsolescence	backward compatibility strategy
Geopolitical Risk	Changes in international landscape may affect global acceptance of the standard	Maintain technology-neutral positioning, de-emphasize geopolitical coloring, focus on solving real problems as the core driving force

12. Conclusion · 结语

In the digital age, trust is the most precious public good.

From the Industrial Revolution to the Information Revolution, every leap in human society has been accompanied by iterations in trust mechanisms. Today, as digital technology reshapes global trade and cultural exchange at an unprecedented pace, we need a more inclusive and pluralistic digital trust infrastructure — one that not only serves efficiency optimization for industrial products but also safeguards the diversity of human culture, ensuring that the unique value of every civilization can be seen, recognized, and respected in the digital world.

The ICO standards framework is committed to providing open, neutral, and inclusive digital trust infrastructure for global cultural and quality assets. We are not building a closed standards system; rather, we are contributing to the global digital trust edifice — using W3C’s technical language to write chapters of cultural value; using multi-stakeholder governance wisdom to build cross-cultural bridges of trust.

We look forward to a future where every product carrying human wisdom and cultural value possesses a trustworthy digital identity — making global trade fairer, cultural heritage more vibrant, and the digital economy more human-centered.

We invite stakeholders worldwide to submit revision proposals by August 20, 2026 (Accelerated Procedure · 45-day review period), and to join us in building a more inclusive global digital trust future.

Annex A (Normative): DPP-CQ Technical Specification Draft

附录A（规范性）DPP-CQ技术规格草案

This annex presents a summary of the DPP-CQ standard technical specification draft. The complete technical specification will be published in the official version of ICO Std 2001-2026.

A.1 DID Identifier Format

The DPP-CQ product DID uses the `did:ico:dpp` method name, with the following format:

```
did:ico:dpp:<namespace>:<unique-id>
```

- `namespace` : Category or issuer namespace (e.g., `tea` , `ceramic` , `fashion`)
- `unique-id` : Product unique identifier; UUID v4 or GS1-like coding is recommended

Examples:

```
did:ico:dpp:tea:longjing-xh-2026-001
did:ico:dpp:ceramic:jz-yd-2026-0892
```

A.2 JSON-LD Context Example

The JSON-LD context definition for DPP-CQ credentials:

```

{
  "@context": [
    "https://www.w3.org/2018/credentials/v1",
    "https://www.w3.org/2018/credentials/examples/v1",
    "https://icoun.org/contexts/dpp-cq/v1"
  ],
  "id": "did:ico:dpp:tea:longjing-xh-2026-001",
  "type": ["VerifiableCredential", "DPPQualityCredential", "CulturalHeritageCredential"],
  "issuer": "did:ico:issuer:zjtea-assoc",
  "issuanceDate": "2026-07-15T00:00:00Z",
  "expirationDate": "2027-07-14T23:59:59Z",
  "credentialSubject": {
    "id": "did:ico:dpp:tea:longjing-xh-2026-001",
    "productName": "西湖龙井茶",
    "category": "geographical-indication-tea",
    "qualityGrade": "Premium",
    "origin": {
      "type": "GeographicalIndication",
      "giNumber": "GI-CN-0001",
      "region": "杭州西湖产区",
      "country": "CN"
    },
    "culturalData": {
      "heritageStatus": "National Intangible Cultural Heritage",
      "heritageYear": 2008,
      "craftMethod": "手工炒制 · 十大手法",
      "artisanInfo": {
        "name": "(可选, 经授权披露)",
        "generation": "第5代传承人"
      }
    },
    "traceability": {
      "harvestDate": "2026-03-28",
      "processLocation": "杭州龙坞茶镇",
      "batchNumber": "XH20260328A01"
    },
    "qualityAttributes": {
      "appearance": "扁平光滑, 色泽嫩绿",
      "aroma": "嫩香持久",
      "taste": "鲜醇甘爽"
    }
  },
  "proof": {
    "type": "BbsBlsSignature2020",
    "created": "2026-07-15T10:30:00Z",
    "proofPurpose": "assertionMethod",
    "verificationMethod": "did:ico:issuer:zjtea-assoc#keys-1",
    "proofValue": "..."
  }
}

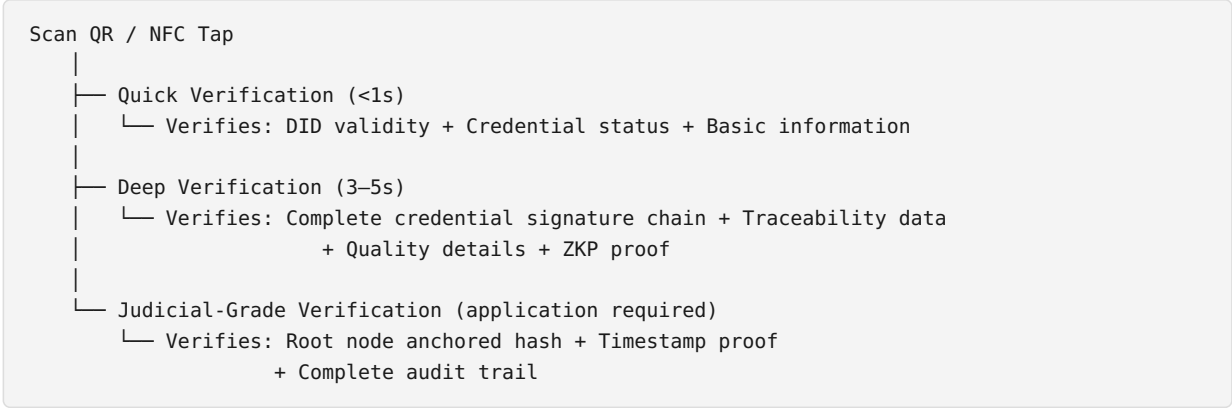
```

A.3 Root Resolver Node API Specification (Draft)

Endpoint	Method	Function
/v1/did/{did}	GET	Query DID document
/v1/credential/{credentialId}/status	GET	Query credential status (valid/revoked/expired)
/v1/credential/verify	POST	Verify credential signature and validity

/v1/register	POST	Register new credential hash
/v1/revoke	POST	Revoke credential
/v1/proof/{hash}	GET	Obtain hash anchoring proof

A.4 Three-Tier Verification Process



Annex B (Informative): Glossary

附录B（资料性）术语表

Term	Definition
Digital Product Passport (DPP)	A digital identity credential containing core product data, readable via methods such as code scanning
Decentralized Identifier (DID)	A decentralized identity identifier defined by W3C, independent of centralized identity providers
Verifiable Credential (VC)	A digital credential digitally signed by the issuer, independently verifiable by third parties
Selective Disclosure (SD-JWT / BBS+)	Cryptographic proofs (SD-JWT per RFC 9529; BBS+ unlinkable derived credentials) that let a holder reveal only selected claims while keeping the signature valid; v1.x used the umbrella term “zero-knowledge proofs (ZKP)”, v2.0 normatively uses these standard suites
Selective Disclosure	The practice by which a data holder discloses only the necessary data fields to a verifier
Geographical Indication (GI)	An indication identifying a product as originating from a specific territory, where a given quality, reputation, or other characteristic of the product is essentially attributable to its geographical origin
Intangible Cultural Heritage (ICH)	Practices and expressions recognized by communities, groups, and, in some cases, individuals as part of their cultural heritage, as defined by UNESCO
Root Resolver	A unified resolution and anchoring infrastructure for globally distributed identifiers
Hash Anchoring	Storing hash fingerprints of data on tamper-proof media for integrity verification and timestamping
	JSON for Linked Data — a JSON format for linked data,

JSON-LD	supporting semantic interoperability
Multi-Stakeholder Governance	A governance model ensuring that all parties affected by decisions have the opportunity to participate in decision-making

Annex C (Informative): Abbreviations

附录C（资料性）缩略语

Abbreviation	Full Form
ICO	International Communication Organization 国际传播组织
DPP	Digital Product Passport 数字产品护照
DPP-CQ	Digital Product Passport for Cultural & Quality Goods 文化与品质数字产品护照
DID	Decentralized Identifier 分布式标识符
VC	Verifiable Credential 可验证凭证
ZKP	Zero-Knowledge Proof 零知识证明
GI	Geographical Indication 地理标志
ICH	Intangible Cultural Heritage 非物质文化遗产
GDCC	Global Digital Credibility Assessment 全球数字公信力评定
TCR	Tianji Credibility Registry 天机可信价值评级数据库
GDC	Global Digital Compact 全球数字契约
W3C	World Wide Web Consortium 万维网联盟
WIPO	World Intellectual Property Organization 世界知识产权组织
UNESCO	United Nations Educational, Scientific and Cultural Organization 联合国教科文组织
ISO	International Organization for Standardization 国际标准化组织
IETF	Internet Engineering Task Force 互联网工程任务组
UNECE	United Nations Economic Commission for Europe 联合国欧洲经济委员会
ESPR	Ecodesign for Sustainable Products Regulation 可持续产品生态设计法规
GS1	Global Standard 1 国际物品编码组织
GDPR	General Data Protection Regulation 通用数据保护条例
UNCITRAL	United Nations Commission on International Trade Law 联合国国际贸易法委员会
NFC	Near Field Communication 近场通信
CLA	Contributor License Agreement 贡献者许可协议

Annex D (Informative): References

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Annex E (Informative): Frequently Asked Questions

附录E（资料性）常见问题

Q1: What is ICO? What qualifications does it have to develop international standards?

ICO (International Communication Organization) is a multi-stakeholder international cooperation platform dedicated to advancing global digital inclusion and cross-cultural digital trust.

The ICO standards framework employs an open community-driven development model, following best practices in international standard-setting (openness, transparency, consensus, balanced representation). The vitality of standards lies in adoption and ecosystem, not administrative mandate — the core Internet protocols (TCP/IP, HTTP, etc.) were not promulgated by governments or official bodies, but became de facto standards through joint advancement by open-source communities and standards organizations and widespread market adoption. ICO is committed to becoming the de facto standard-setter in the field of cultural and quality digital trust.

Q2: What is the relationship between ICO standards and the EU DPP? Will they conflict?

A complementary relationship, not a competitive one.

The EU DPP (under the ESPR framework) focuses primarily on environmental and circular economy data for industrial products — a “green dimension” Digital Product Passport. ICO DPP-CQ focuses on the cultural value

and quality dimensions of cultural products and specialty agricultural goods — a “cultural dimension” Digital Product Passport. The two cover different product categories and data dimensions; they are complementary rather than substitutive.

At the technical level, DPP-CQ is fully compatible with W3C DID/VC standards, sharing the same technical foundation as the EU DPP, enabling data interoperability. v2.0 makes this concrete: products in the GS1 interop profile carry GTIN identifiers and open GS1 Digital Link carriers (aligned with EN 18219/18220), credentials support SD-JWT VC (RFC 9529) alongside BBS+, and sustainability data follows ISO 14067 and UNTP-style conformity claims, so EU importers and regulators can read and cross-check DPP-CQ data with standard tooling. DPP-CQ’s flagship categories (tea, wine & spirits, cultural crafts, traditional medicine products) are not in the first ESPR priority product groups, so there is no statutory conflict; the two regimes address different dimensions and can be carried by the same passport. We welcome standard coordination and mutual recognition dialogue with relevant EU institutions.

Q3: Why is a Root Resolver Network needed? Will it create a monopoly?

The core function of the Root Resolver Network is to ensure consistency of DID resolution and queryability of credential status in global cross-border circulation. This is analogous to the root servers of the Domain Name System (DNS) — not for the purpose of monopolization, but for the purpose of interoperability.

The design of the root resolution mechanism follows the following principles: - **Technology-neutral**: Root nodes only provide resolution and anchoring services; they do not participate in standard-setting and do not engage in commercial competition; - **Governance constraints**: Root node operation is overseen by the Technical Committee and authorized by the Standards Council; major changes require approval through governance processes; - **Gradual decentralization**: The long-term goal is to establish a distributed root node network with multiple regions and multiple operators, avoiding single-point dependencies; - **Open source and open**: The reference implementation of root nodes is fully open-source, and any institution may deploy a compatible node.

Q4: How are data security and privacy ensured? Will data exit issues arise?

ICO standards employ an architectural design of “data localization, hash cross-border”: - **Raw data stored locally**: Detailed product quality, craftsmanship, and supply chain data are stored locally by the issuer or certification body, with no mandatory cross-border transmission; - **Hash fingerprint anchoring**: Only hash fingerprints of data (one-way irreversible) are registered and anchored at root nodes; hashes themselves contain no identifiable original information; - **Selective disclosure**: Sensitive data is verified via SD-JWT / BBS+ proofs; verifiers receive only the claims or compliance conclusions they are authorized for, without access to undisclosed raw data; - **Compliance adaptation**: Regional nodes may conduct localized deployment and compliance adjustments in accordance with local data regulations.

This architecture fundamentally reduces the risk of cross-border flow of raw data and complies with regulatory requirements for data sovereignty and data security in all countries.

Q5: What is the relationship between the Tianji system and ICO?

The Tianji system (天机体系) is one of the institutions undertaking the role of Technical Secretariat for the ICO standards framework, responsible for reference implementation of standards, root node operation and maintenance, and technical support.

This arrangement is analogous to the relationship between W3C and host institutions such as MIT/ERCIM — standards are developed jointly by the community, and the Technical Secretariat is responsible for execution and implementation. The Tianji system, as a technical service provider, does not participate in standard voting, ensuring separation of standard-setting from technical execution.

Q6: Is there a fee to adopt ICO standards?

The standards themselves are completely free. - Standards documentation: Free to download and use (CC BY 4.0 / Apache 2.0) - Reference implementations: Open-source and free, freely modifiable and commercially usable - Basic verification: Basic functions such as scan-to-verify are freely available

Value-added services (such as custom development, advanced assessment services, technical support, etc.) may be provided by service providers in the ecosystem; ICO itself does not directly operate commercial services.

Q7: What is the relationship with GS1 and ISO? Is this a separate stovepipe effort?

No, it is not a separate stovepipe effort. ICO standards are built upon existing international standards such as W3C, ISO, and GS1, serving as a supplement and extension of the existing standards ecosystem in the cultural and quality dimensions.

W3C DID is the “technical standard for ID cards” (telling us how to make the card), ISO/GS1 is the “industry coding specification” (telling us how to number things), and ICO DPP-CQ is the “content specification for cultural product passports” (telling us what content goes in the passport and how to verify cultural value). We use W3C’s card technology, are compatible with GS1’s coding system, and focus on content standards for the vertical domain of culture and quality.

We actively seek cooperation and mutual recognition with standards organizations such as ISO, W3C, and GS1.

Q8: Can small and medium-sized producers afford it? Will it increase their burden?

This is precisely one of the core concerns of ICO standards.

Our design goal is to enable small and medium-sized producers and ICH transmitters to “afford and benefit from” the system: - **Low-cost solutions:** Support low-cost verification methods such as QR codes, with startup costs as low as a few cents per item; - **Lightweight onboarding:** SaaS-based assessment platforms are provided, requiring no technical team — usable upon registration; - **Volume discounts:** Discounts or waivers for small and medium-sized producers and ICH transmitters; - **Capacity building:** Training and technical support provided through regional nodes and partners.

We believe that the “premium for quality” effect brought by digital passports will far exceed their usage cost — helping good products command good prices is the most direct value empowerment for producers.

Q9: What about the legality and evidentiary effect of the standards?

ICO standards are themselves a set of technical and industry standards, without legal force (consistent with the nature of international standards such as those from ISO). However, their design fully considers judicial evidence requirements:

- **Hash anchoring:** Cryptography-based data integrity proof can serve as electronic evidence in most jurisdictions;
- **Trusted timestamps:** Integration with authoritative timestamp services meets the temporal certainty requirements for electronic evidence;
- **Complete evidence chain:** Issuance chains, change records, and revocation logs constitute a complete evidence chain;
- **Third-party audits:** Root nodes and core services undergo regular independent third-party audits to enhance evidentiary credibility.

Specific legal effect varies by jurisdiction. ICO encourages regional nodes to collaborate with local legal institutions to promote recognition of the evidentiary weight of DPP-CQ credentials in their respective jurisdictions.

Q10: How is the neutrality of the standards ensured? How can we be certain no single party dominates?

The multi-stakeholder governance mechanism is the institutional guarantee for standard neutrality:

- **Balanced representation:** Council seats are allocated by category (government 25%, industry 25%, academia 20%, civil society 20%, Technical Secretariat 10% non-voting), so no single group can dominate decision-making;
- **Consensus-first:** Important decisions prioritize seeking consensus rather than simple majority voting;
- **Separation of technology and standards:** The Technical Secretariat does not participate in standard voting, separating execution from decision-making;
- **Transparency and openness:** Governance processes are open and transparent, subject to community oversight;
- **Open participation:** Any institution may apply to join, regardless of nationality, size, or ownership structure.

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Annex F (Normative): v2.0 Interoperability & Compliance Annexes

附录F（规范性）v2.0 互操作与合规附件

The following separate documents form a normative part of ICO Std 2001 v2.0. They are maintained in the repository under docs/：

Document	Path	Content
GS1 Digital Link & Identifier Mapping	<code>docs/interoperability/gs1-digital-link-mapping.md</code>	GTIN↔DID binding rules, Digital Link URI structure and content negotiation, EPCIS 2.0 event mapping, EN 18219/18220 alignment
Data Carrier Specification	<code>docs/interoperability/carrier-specification.md</code>	Three-tier carriers (L1 open QR / L2 SDM NFC / L3 tamper-evident NFC), open-carrier principle, dual-carrier rule, resolver behavior
Credential Format Profiles	<code>docs/specs/credential-formats.md</code>	JSON-LD Data Integrity (BBS+) vs SD-JWT VC (RFC 9529) profiles; disclosure classification; SM2/SM3/SM4 national cryptography suite; status and validity rules
Conformity Declarations	<code>docs/compliance/conformity-declarations.md</code>	Self-declarations against GB/T 47507-2026, EU ESPR/EN standards, GDPR/PIPL, EU AI Act
PIA / DPIA Summary	<code>docs/compliance/pia-summary.md</code>	Public privacy impact assessment: data flows, risk table, data-subject rights, cross-border legal basis
JSON Schema v2.0	<code>schemas/dpp-cq.schema.json</code>	Machine-readable data model (\$id: icoun.org/schemas/dpp-cq/v2.0.0-draft)

v2.0 Example Credential	examples/longjing-tea-v2.json	Full interop-profile example (Longjing tea)
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Relationship to external frameworks

External framework	Nature	DPP-CQ posture
W3C DID / VC v2.0, IETF SD-JWT (RFC 9529)	Open standards	Foundation; fully adopted
GS1 Digital Link / EPCIS 2.0	Global supply-chain standards	Normative interop profile
EU ESPR (Reg. 2024/1781); EN 18219–18223	EU regulation / harmonized standards	Baseline compatibility for carriers, identifiers, sustainability fields; DPP-CQ categories not in first priority groups
UN/CEFACT UNTP	UN multilateral vocabulary	Adopted for conformity claims; primary multilateral bridge
ISO 14067, ISO 22000, ISO 22739	ISO standards	Referenced in data model and terminology
GB/T 47507-2026; SM2/SM3/SM4 (GB/T 32918/32905/32907)	Chinese national standards	Supported for domestic deployments
EU AI Act (Reg. 2024/1689) Art. 50	EU regulation	Transparency provisions adopted as baseline practice

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This is a public review draft. Content is subject to change based on community feedback. The official version published by ICO shall prevail.

本文档为征求意见稿，内容可能根据社区反馈进行修改。正式版本以ICO官方发布为准。

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Normative Annexes (v2.0)

The following documents form a normative part of ICO Std 2001 v2.0.

Annex G: GS1 Digital Link Mapping

Normative Annex — ICO Std 2001 v2.0

Status: Normative · Part of ICO Std 2001:2026 (DPP-CQ v2.0)

Maintained at: docs/interoperability/gs1-digital-link-mapping.md

Version: 2.0.0-draft

1. Scope

This annex defines how DPP-CQ credentials interoperate with the GS1 identification and Digital Link ecosystem. It is mandatory for all credentials issued under the GS1 interop profile described in the white paper (§8.2b).

2. Identifier Binding Rules

2.1 GTIN ↔ DID Binding

Every DPP-CQ credential in the GS1 interop profile **MUST** carry both a GS1 Global Trade Item Number (GTIN) and a DPP-CQ Decentralized Identifier (DID).

Identifier	Role	Format	Governance
GTIN-14	Trade item identification (global supply chain)	14-digit numeric (GS1 General Specifications)	GS1 Member Organisation
DID	Credential identity & resolution	did:ico:dpp:<namespace>:<unique-id>	ICO

Binding rules:

1. A single GTIN **MAY** map to multiple DIDs (e.g., per-batch credentials for the same trade item).
2. Each DID **MUST** resolve to exactly one GTIN via the `gs1DigitalLink` property in the credential.
3. The binding **MUST** be recorded in the credential's `gs1DigitalLink` object:

```
{
  "gs1DigitalLink": {
    "gtin": "06141411234567",
    "did": "did:ico:dpp:tea:longjing-xh-2026-001",
    "identifierRelation": "one-to-many"
  }
}
```

2.2 DID Format

The DPP-CQ DID uses the `did:ico:dpp` method name:

```
did:ico:dpp:<namespace>:<unique-id>
```

- `namespace` : Product category namespace (e.g., `tea` , `wine` , `ceramic`)
- `unique-id` : Product unique identifier; UUID v4 recommended for global uniqueness, or GS1-compatible coding for supply-chain alignment

2.3 Multi-Issuer Scenarios

When multiple entities issue credentials for the same product (e.g., producer DPP + certifier quality credential), each issuer generates its own DID and credential. The `gs1DigitalLink.gtin` field links them to the same trade item.

3. Digital Link URI Structure

3.1 URI Template

The GS1 Digital Link URI **MUST** follow this structure:

```
https://<resolver-domain>/{GTIN}
```

Where `<resolver-domain>` is the issuer's or a third-party GS1 Digital Link resolver (e.g., `https://id.gs1.org/01/06141411234567`).

3.2 Content Negotiation

The resolver **MUST** support content negotiation per the GS1 Digital Link standard (GS1 GS1 2.2):

Accept Header	Response
<code>text/html</code>	Human-readable product information page (verification result)
<code>application/ld+json</code>	JSON-LD representation of the DPP-CQ credential
<code>application/vc+ld+json</code>	Verifiable Credential (JSON-LD, BBS+ proof)
<code>application/vc+sd-jwt</code>	SD-JWT VC representation
<code>application/json</code>	Simplified JSON summary

3.3 Resolver Behavior

1. **Resolution:** The resolver receives a Digital Link URI, extracts the GTIN, and queries the ICO resolver API for the associated DID and credential hash.
2. **Presentation:** Based on the `Accept` header, the resolver returns the appropriate representation.
3. **Verification:** The resolver **MUST** verify credential validity (status check against the status list) before returning the credential data.
4. **Caching:** Resolved credentials **MAY** be cached for up to 24 hours; invalidation **MUST** occur upon revocation.

3.4 Link Header (RFC 8288)

The resolver **SHOULD** include `Link` headers pointing to related resources:

```
Link: <https://resolver.icoun.org/v1/did/did:ico:dpp:tea:longjing-xh-2026-001>; rel="canonical"
Link: <https://resolver.icoun.org/v1/credential/status>; rel="status"
```

4. EPCIS 2.0 Event Mapping

DPP-CQ credentials **MAY** be complemented by EPCIS 2.0 (Electronic Product Code Information Services) events for granular supply chain traceability.

4.1 Event Types

EPCIS Event	DPP-CQ Mapping	Notes
ObjectEvent	<code>credentialSubject.traceability.supplyChainSteps[]</code>	Harvest, processing, packaging
AggregationEvent	<code>credentialSubject.traceability.aggregation</code>	Pallet/case grouping
TransactionEvent	<code>credentialSubject.traceability.transaction</code>	Sale, transfer of ownership
TransformationEvent	<code>credentialSubject.traceability.transformation</code>	Processing that changes identity

4.2 Event Structure

EPCIS events **MUST** reference the DPP-CQ credential DID in the `epcList` or `childEPCs` field:

```
{
  "type": "ObjectEvent",
  "action": "OBSERVE",
  "bizStep": "commissioning",
  "epcList": ["did:ico:dpp:tea:longjing-xh-2026-001"],
  "eventTime": "2026-03-28T06:00:00Z",
  "eventTimeZoneOffset": "+08:00",
  "readPoint": { "id": "urn:epc:id:sgln:0614141.00001.0" }
}
```

4.3 Storage & Access

- EPCIS events **SHOULD** be stored in a GS1-compliant EPCIS repository.
- The repository endpoint **MAY** be referenced in the credential via `credentialSubject.traceability.epcisEndpoint`.
- Access control **MUST** follow the same selective-disclosure principles as the credential itself (verifier receives only authorized events).

5. EN 18219 / EN 18220 Alignment

This annex aligns with the European standard series for digital product passports:

Standard	Scope	DPP-CQ Alignment
EN 18219	Data carriers for DPP — technical requirements	§3 Digital Link URI structure; open-carrier principle (§7.7 of white paper)
EN 18220	DPP data model — core data attributes	§2 identifier binding; §4 EPCIS mapping; dual-dimension model (quality + sustainability)

5.1 Compliance Posture

DPP-CQ adopts a “**baseline compatibility, differentiated value**” posture:

- **Baseline compatibility (non-negotiable):** open data carriers, global identifiers (GTIN alongside DID), standard credential formats (W3C VC / SD-JWT VC), standard APIs, and machine-readable conformity claims.
- **Differentiated value (independent):** quality-and-culture assessment methodology (ICO Std 3001), multi-stakeholder governance, affordability for small producers, and applicability to cultural product categories that environmental DPPs do not cover.

6. Multilateral Bridge: UN/CEFACT UNTP

DPP-CQ’s conformity claims vocabulary is aligned with the UN Transparency Protocol (UNTP) to enable cross-jurisdiction interoperability through neutral, multilateral standards rather than any single jurisdiction’s registry.

UNTP Concept	DPP-CQ Mapping
conformityClaim (UNTP)	credentialSubject.qualityAttributes.certifications[]
facility (UNTP)	credentialSubject.traceability.supplyChainSteps[].location
product (UNTP)	credentialSubject (product-level credential)
process (UNTP)	credentialSubject.traceability.transformation

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Annex H: Carrier Specification

Normative Annex — ICO Std 2001 v2.0

Status: Normative · Part of ICO Std 2001:2026 (DPP-CQ v2.0) **Maintained at:** docs/interoperability/carrier-specification.md **Version:** 2.0.0-draft

1. Scope

This annex specifies the three-tier physical carrier system for DPP-CQ credentials. It defines carrier levels, security profiles, reader requirements, and the principles governing carrier selection. Aligned with white paper §7.7.

2. Three-Tier Carrier Model

Level	Carrier	Security Profile	Reader Requirement	Clone Resistance
L1	Open QR Code (GS1 Digital Link in interop profile)	Cloneable; authenticity via registry verification	Any smartphone camera	None (registry-dependent)
L2	SDM Secure NFC (NTAG 424 DNA class)	Per-scan cryptographic SUN authentication	Any NFC-capable phone, NDEF open read	High (hardware-enforced)
L3	Tamper-Evident NFC (TagTamper class)	Tamper detection + SDM + physical evidence	Any NFC-capable phone	Very High (tamper-evident)

3. Level Definitions

3.1 L1 — Open QR Code

Definition: A printed QR code encoding a GS1 Digital Link URI that resolves to the DPP-CQ credential.

Requirements: - **Encoding:** GS1 Digital Link URI per [docs/interoperability/gsl-digital-link-mapping.md](#) - **Size:** Minimum 25mm × 25mm for reliable scanning; recommended 30mm × 30mm - **Error correction:** Level M (15%) minimum; Level H (30%) recommended for harsh environments - **Quiet zone:** Minimum 4 modules on all sides - **Print quality:** ISO/IEC 15415 Grade C (2.0) minimum; Grade B recommended

Security model: The QR code is inherently cloneable. Authenticity is established by resolving the URI and verifying the credential's cryptographic signature against the issuer's DID document. No security through obscurity.

Use cases: - Mass-market products where cost is a primary constraint - Products where NFC is impractical (flexible packaging, small items) - Consumer-facing verification (any smartphone can scan)

3.2 L2 — SDM Secure NFC

Definition: An NFC tag (NTAG 424 DNA or equivalent) with Secure Direct Message (SDM) capability, providing per-scan cryptographic authentication.

Requirements: - **Chip class:** NXP NTAG 424 DNA (or equivalent with ISO 18092 compliance) - **UID:** Unique, factory-set 7-byte UID (read-only) - **SDM:** Secure Direct Message with AES-128 encryption - **SUN** (Secure Unique NFC) message generated per scan - Includes tag UID, scan counter, and AES-CMAC authentication - **Memory:** Minimum 1024 bytes user memory for credential metadata - **NDEF format:** NDEF message **MUST** contain the GS1 Digital Link URI as the primary payload; **SHOULD** also include an SDM mirror URL for authentication

Security model: Each NFC tap generates a unique cryptographic response (SUN message) that proves the tag is genuine and has not been cloned. The verification backend validates the AES-CMAC against the tag's secret key.

Use cases: - Premium products requiring anti-counterfeiting (wine, spirits, luxury goods) - Products sold in environments where QR scanning is inconvenient - Dual-carrier configurations (see §5)

3.3 L3 — Tamper-Evident NFC

Definition: An NFC tag with integrated tamper-detection capability, providing both cryptographic authentication and physical evidence of tampering.

Requirements: - **Chip class:** NTAG 424 DNA with TagTamper feature, or equivalent (e.g., NXP NTAG 424 DNA TT) - **Tamper detection:** Conductive trace that breaks upon removal; generates a tamper-evident flag in the SDM response - **Tamper response:** SDM response **MUST** include tamper status bit; verification page **MUST** display warning if tamper flag is set - **Adhesive:** Tamper-evident destructible label material; removal causes irreversible physical damage

Security model: Combines L2's cryptographic authentication with physical tamper evidence. If the tag is removed from its original surface, the conductive trace breaks, and subsequent scans report tampered status.

Use cases: - High-value products (precious metals, luxury watches, rare teas) - Regulated products requiring chain-of-custody evidence - Products where label swapping is a known risk

4. Open Carrier Principle

Aligned with EU ESPR Art. 9 and EN 18220, DPP-CQ mandates the **open carrier principle**:

*Every product **MUST** be readable with generally available means — no proprietary app, registration, or fee.
Security is achieved through cryptographic verification of the resolved data, never through hiding the payload.*

4.1 Requirements

1. **No proprietary apps**: The QR code or NFC tag **MUST** be readable with the device's native camera or NFC reader. No specialized app required.
2. **No registration wall**: Accessing the verification page **MUST NOT** require user registration or account creation.
3. **No fee**: Accessing basic verification (authenticity + core credential data) **MUST** be free of charge.
4. **Open standards**: All data formats **MUST** be based on open standards (W3C VC, GS1 Digital Link, NDEF).

4.2 Verification Page

The verification page (resolved via the Digital Link URI) **MUST**: - Display product name, quality grade, and issuer information - Show credential validity status (valid / revoked / expired) - Indicate which carrier was presented (QR / NFC / Tamper-evident) - Provide links to full credential data and conformity claims - Be accessible without JavaScript (basic functionality via server-rendered HTML)

5. Dual Carrier Rule

For products in the GS1 interop profile, the **dual carrier rule** applies:

*Products **SHOULD** carry both an L1 open QR code (universal scanning, retail/customer flows) and an L2 SDM NFC tag (cryptographic authenticity); both **MUST** resolve to the same passport, and the verification page **MUST** indicate which carrier was presented.*

5.1 Configuration

Configuration	When to Use
L1 only	Cost-constrained; low-counterfeit-risk products
L1 + L2	Standard for GS1 interop profile; most consumer products
L2 only	Products where QR is impractical; NFC-only packaging
L3 only	High-value; tamper-evident required
L1 + L2 + L3	Premium tier; QR for retail, NFC for authentication, tamper evidence for chain-of-custody

5.2 Carrier Registration

The carrier tier configuration **MUST** be recorded in the credential:

```
{
  "carrierTier": {
    "level": "L2",
    "primary": "sdm-nfc",
    "secondary": "open-qr",
    "nfcChipType": "NTAG-424-DNA",
    "tamperEvident": false,
    "dualCarrier": true
  }
}
```

6. Resolver Behavior

The ICO resolver **MUST** handle carrier-specific behavior:

Carrier	Resolution	Authentication
L1 (QR)	Resolve Digital Link URI → fetch credential → verify signature	Credential-level only
L2 (SDM NFC)	Parse SUN message → validate AES-CMAC → fetch credential → verify signature	Tag-level + credential-level
L3 (Tamper NFC)	Parse SUN message → check tamper flag → validate AES-CMAC → fetch credential → verify signature	Tamper check + tag-level + credential-level

6.1 Response Codes

Scenario	Response
Valid credential, genuine tag	200 OK — full verification result
Valid credential, tamper flag set (L3)	200 OK — verification result + tamper warning
Revoked credential	200 OK — status: revoked, with details
Invalid SUN message (L2/L3)	401 Unauthorized — tag authentication failed
Unknown DID	404 Not Found

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Annex I: Credential Formats

Normative Annex — ICO Std 2001 v2.0

Status: Normative · Part of ICO Std 2001:2026 (DPP-CQ v2.0) Maintained at: <code>docs/specs/credential-formats.md</code> Version: 2.0.0-draft
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1. Scope

This annex defines the two supported credential format profiles for DPP-CQ v2.0: JSON-LD Data Integrity (BBS+) and SD-JWT VC. It also defines the selective disclosure classification scheme and the optional Chinese national cryptography suite.

2. Format Profiles

2.1 JSON-LD Data Integrity (BBS+)

Specification: W3C Verifiable Credentials Data Model v2.0 + Data Integrity (BBS+ signature suite)

Use case: Selective disclosure of individual claims (field-level granularity); unlinkable presentations.

Profile Requirements

Property	Requirement
@context	MUST include <code>https://www.w3.org/ns/credentials/v2</code> and <code>https://icoun.org/contexts/dpp-cq/v1</code>
type	MUST include <code>VerifiableCredential</code> + at least one DPP-CQ type
proof.type	<code>DataIntegrityProof</code> (VC 2.0) or <code>BbsBlsSignature2020</code> (legacy)
proof.cryptosuite	<code>bbs-2023</code> (VC 2.0 Data Integrity)
proof.proofPurpose	<code>assertionMethod</code>
proof.verificationMethod	DID URL pointing to issuer's BBS+ public key

Selective Disclosure

BBS+ enables the holder to derive a presentation containing only selected claims. The verifier can verify the derived presentation without contacting the issuer.

Disclosure classification (see §3 for full scheme): - **Public claims:** Always disclosed (product name, grade, origin) - **Holder-selectable claims:** Disclosed at holder's discretion (artisan info, detailed specifications) - **Restricted claims:** Disclosed only to authorized verifiers (full supply chain, batch-level data)

Example (VC 2.0 Data Integrity)

```
{
  "@context": [
    "https://www.w3.org/ns/credentials/v2",
    "https://icoun.org/contexts/dpp-cq/v1"
  ],
  "id": "did:ico:dpp:tea:longjing-xh-2026-001",
  "type": ["VerifiableCredential", "DPPQualityCredential"],
  "issuer": "did:ico:issuer:zjtea-assoc",
  "validFrom": "2026-07-15T00:00:00Z",
  "validUntil": "2027-07-14T23:59:59Z",
  "credentialSubject": { ... },
  "proof": {
    "type": "DataIntegrityProof",
    "cryptosuite": "bbs-2023",
    "created": "2026-07-15T10:30:00Z",
    "proofPurpose": "assertionMethod",
    "verificationMethod": "did:ico:issuer:zjtea-assoc#keys-bbs-1",
    "proofValue": "u2V0..."
  }
}
```

2.2 SD-JWT VC (IETF RFC 9529)

Specification: IETF SD-JWT VC (RFC 9529) — Selective Disclosure for JWTs

Use case: Efficient credential presentation in mobile wallets; compact serialization; widely supported by mobile platform wallets (Apple Wallet, Google Wallet).

Profile Requirements

Property	Requirement
Header typ	vc+sd-jwt
Header alg	ES256 (ECDSA P-256) mandatory; ES384 optional
Payload vct	https://icoun.org/vct/dpp-cq/v1 (Verifiable Credential Type)
Payload iss	Issuer DID or HTTPS URL
Payload cnf	Holder binding (JWK or DID)
Payload sd	Array of salted digests for selectively disclosable claims
_sd_alg	sha-256 (default) or sha-384

SD-JWT Structure

```
<issuer-signed JWT>~<disclosure 1>~<disclosure 2>~...~<disclosure N>~<key binding JWT, optional>
```

Each disclosure is a base64url-encoded JSON array: [salt, claim_name, claim_value]

Example Disclosure

For the claim "qualityGrade": "Premium Grade 1" :

```
WyI5MjNkMzQ1NiIsICJxdWFsaXR5R3JhZGUlLCAiUHJlbWl1bSBHcmFkZSAxIl0
```

2.3 Format Selection Guidance

Criterion	BBS+ (JSON-LD)	SD-JWT VC
Granularity	Field-level selective disclosure	Claim-level selective disclosure
Unlinkability	Unlinkable derived presentations	⚠ Issuer-signature JWT (linkable if not carefully managed)
Compactness	Larger (JSON-LD overhead)	Compact (JWT serialization)
Mobile wallet support	Limited (emerging)	Broad (Apple/Google Wallet)
Semantic interoperability	JSON-LD (RDF-based)	JWT claims (flat structure)
Recommended for	Regulatory / multi-party verification	Consumer-facing mobile flows

Interoperability: Both formats resolve to the same semantic data model (defined in schemas/dpp-cq.schema.json). A resolver MUST be able to serve both formats for the same credential.

3. Disclosure Classification

DPP-CQ defines three disclosure classes for credential claims:

Class	Label	Disclosure Rule	Examples
P	Public	Always disclosed in any presentation	Product name, category, quality grade, origin country, credential status
		Disclosed at holder's discretion	Artisan name, detailed

H	Holder-selectable	Disclosed at holder's discretion per presentation	specifications, cultural narrative, supply chain steps
R	Restricted	Disclosed only to authorized verifiers (e.g., regulators, customs)	Batch-level data, full supply chain, laboratory results, personal data

3.1 Default Classification

The following table provides default classifications. Issuers MAY elevate (but not lower) the classification of specific claims:

Claim	Default Class	Notes
productName	P	Core identification
category	P	Core identification
qualityGrade	P	Core quality claim
origin.country	P	Core origin
origin.giNumber	P	GI registration is public
origin.region	P	Core origin
origin.coordinates	H	May reveal precise location
culturalData.heritageStatus	P	Public classification
culturalData.artisanInfo.name	R	Personal data; requires consent
culturalData.culturalNarrative	H	Optional narrative
traceability.batchNumber	R	Batch-level operational data
traceability.supplyChainSteps	H	Holder decides granularity
qualityAttributes.specifications	H	Lab results may be competitive
qualityAttributes.certifications	P	Certifications are public record
sustainability.*	H	Holder decides disclosure scope
aiDisclosure.*	P	Transparency is mandatory (see §4)
dataLifecycle.*	P	Data governance transparency

3.2 AI Disclosure (Mandatory)

Where AI systems support quality assessment, the credential **MUST** include:

```

{
  "qualityAttributes": {
    "assessment": {
      "aiDisclosure": {
        "aiUsed": true,
        "aiRoles": ["data-extraction", "pre-scoring"],
        "humanOversight": true,
        "finalDecisionBy": "human-assessor",
        "methodologyReference": "https://resolver.icoun.org/methods/tea-premium-v3",
        "reportReference": "https://resolver.icoun.org/reports/XH20260328A01",
        "regulatoryBasis": ["GB/T 47507-2026", "EU AI Act Art.50"]
      }
    }
  }
}

```

```
}
```

Rules: - `aiDisclosure` is **MANDATORY** if AI was used in any assessment step - `aiDisclosure.aiUsed` **MUST** be `true` when AI participated - The final grading decision **MUST** be made by a qualified human assessor - The assessment methodology and full report **MUST** be retrievable via URI

4. Chinese National Cryptography Suite (Optional)

For deployments within mainland China or other jurisdictions requiring national cryptography standards, DPP-CQ supports an optional SM2/SM3/SM4 cryptography suite.

4.1 Algorithm Mapping

Function	International Default	Chinese Suite (Optional)	Standard
Digital Signature	Ed25519 / ES256	SM2	GB/T 32918
Hash Function	SHA-256	SM3	GB/T 32905
Symmetric Encryption	AES-128/256	SM4	GB/T 32907

4.2 Implementation

When the Chinese suite is selected:

1. **Signature:** `proof.type` = `DataIntegrityProof`, `proof.cryptosuite` = `sm2-2024`
2. **Hash:** All hash operations use SM3 instead of SHA-256
3. **Encryption:** All symmetric encryption uses SM4 instead of AES

4.3 Interoperability Note

Credentials signed with the Chinese suite **MUST** still be resolvable via the standard DID document. The issuer's DID document **MUST** list both key types:

```
{
  "verificationMethod": [
    {
      "id": "did:ico:issuer:zjtea-assoc#keys-bbs-1",
      "type": "Bls12381G2Key2020",
      "controller": "did:ico:issuer:zjtea-assoc",
      "publicKeyMultibase": "z..."
    },
    {
      "id": "did:ico:issuer:zjtea-assoc#keys-sm2-1",
      "type": "SM2VerificationKey2024",
      "controller": "did:ico:issuer:zjtea-assoc",
      "publicKeyMultibase": "z..."
    }
  ]
}
```

5. Status & Validity Rules

5.1 Credential Status

Credentials **SHOULD** use the W3C BitstringStatusList (VC 2.0) for status management:

```
{
  "credentialStatus": {
    "id": "https://resolver.icoun.org/v1/status/001#42",
    "type": "BitstringStatusListEntry",
    "statusPurpose": "revocation",
    "statusListIndex": "42",
    "statusListCredential": "https://resolver.icoun.org/v1/status/001"
  }
}
```

5.2 Validity Period

- `validFrom` / `validUntil` (VC 2.0) or `issuanceDate` / `expirationDate` (VC 1.x legacy) **SHOULD** be set for all credentials
- Quality-sensitive products (food, tea) **MUST** have an expiration date
- Expired credentials **MUST** return status `expired` upon verification
- Revoked credentials **MUST** return status `revoked` with revocation reason

5.3 Revocation

- Revocation is performed by setting the corresponding bit in the status list
- Revocation **MUST** be performed by the issuer or an authorized revocation service
- Revoked credentials remain verifiable (signature is still valid) but return status `revoked`
- Revocation reason **SHOULD** be provided in the status list entry

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Annex J: Conformity Declarations

Normative Annex — ICO Std 2001 v2.0

Status: Normative · Part of ICO Std 2001:2026 (DPP-CQ v2.0) **Maintained at:** `docs/compliance/conformity-declarations.md` **Version:** 2.0.0-draft

1. Scope

This annex contains self-declarations of conformity made by ICO against relevant regulatory frameworks and standards. These declarations are maintained as part of the standard and are updated as frameworks evolve.

Aligned with UNTP-style conformity claims (white paper §8.2b).

2. Declaration Framework

Each declaration follows this structure:

Field	Description
Framework	The external regulation or standard
Scope	What aspects of DPP-CQ are covered

Conformity level	Full / Partial / Aligned
Evidence	How conformity is demonstrated
Limitations	Known gaps or areas of non-conformity

3. AI Trustworthiness: GB/T 47507-2026

Framework: GB/T 47507-2026 — Information technology — Artificial intelligence — General rules for trustworthiness of AI systems

GB/T 47507-2026 Requirement	DPP-CQ Implementation	Conformity
AI transparency	Mandatory <code>aiDisclosure</code> in credential (§4 of <code>credential-formats.md</code>)	Full
Human oversight	Final grading decision MUST be made by qualified human assessor	Full
Traceability of AI decisions	<code>methodologyReference</code> and <code>reportReference</code> URIs provide audit trail	Full
Reliability	AI assessment results are one input; human reviewer validates	Full
Privacy	AI processing follows data lifecycle governance (§8.3 of white paper)	Full

Declaration: ICO declares that DPP-CQ v2.0’s AI transparency provisions conform to GB/T 47507-2026’s requirements for AI systems used in quality assessment.

Limitations: This declaration covers the standard’s provisions only. Individual implementations must ensure their specific AI systems comply.

4. EU AI Act (Regulation 2024/1689) — Article 50

Framework: EU Artificial Intelligence Act, Regulation (EU) 2024/1689, Article 50 — Transparency obligations

EU AI Act Art. 50 Requirement	DPP-CQ Implementation	Conformity
AI system disclosure	<code>aiDisclosure.aiUsed</code> mandatory flag	Full
AI role description	<code>aiDisclosure.aiRoles</code> enumerates specific functions	Full
Human oversight statement	<code>aiDisclosure.humanOversight</code> + <code>finalDecisionBy</code>	Full
Content authenticity	Credential provenance chain provides content origin	Aligned

Declaration: ICO declares that DPP-CQ v2.0’s AI transparency provisions are aligned with the direction and intent of EU AI Act Article 50. DPP-CQ exceeds minimum requirements by mandating disclosure (not merely “best effort”) and requiring retrievable methodology references.

Limitations: This declaration covers the standard’s data model provisions. Deployers must assess whether their use of AI in quality assessment triggers additional obligations (e.g., high-risk classification under Art. 6).

5. EU ESPR / EN Standards

Framework: Regulation (EU) 2024/1781 (ESPR) and harmonized standards EN 18219–18223 series

Requirement	DPP-CQ Implementation	Conformity
Open data carrier (ESPR Art. 9)	Open carrier principle (§4 of carrier-specification.md)	Full
No proprietary app required	Open carrier principle, L1 QR readable by any smartphone	Full
Identifier interoperability	GTIN alongside DID; GS1 Digital Link URI	Full
Sustainability data attributes	Optional sustainability module (ISO 14067 PCF)	Aligned
EN 18220 core data attributes	Dual-dimension model; quality + sustainability	Aligned

Declaration: ICO declares that DPP-CQ v2.0's carrier, identifier, and data model provisions are fully aligned with ESPR Art. 9 and baseline-compatible with the EN 18219–18223 series.

Strategic note: DPP-CQ's flagship categories (tea, wine & spirits, cultural crafts, traditional medicine, specialty agri-foods) are not within the first ESPR priority product groups. The standard has a strategic window to mature its interop profile before statutory DPP obligations potentially reach its categories.

6. GDPR (EU) & PIPL (China)

Framework: General Data Protection Regulation (EU) 2016/679; Personal Information Protection Law (China)

Requirement	DPP-CQ Implementation	Conformity
Data minimization	Selective disclosure (BBS+ / SD-JWT); verifiers receive only authorized claims	Full
Purpose limitation	<code>dataLifecycle</code> metadata specifies purpose and retention	Full
Storage limitation	<code>dataLifecycle.retentionPolicy</code> with deletion/anonymization rules	Full
Cross-border transfer	"Data localization, hash cross-border" architecture (§8.1 of white paper)	Full
Data subject rights	<code>dataLifecycle.privacyNoticeUri</code> provides rights information	Full
Privacy by design	Raw data stored locally; only hashes cross borders	Full
Consent management	Artisan personal data classified as Restricted (R) in disclosure classification	Full

Declaration: ICO declares that DPP-CQ v2.0's data architecture and disclosure mechanisms are designed to support conformity with both GDPR and PIPL. The "data localization, hash cross-border" architecture fundamentally reduces cross-border data compliance risks.

Limitations: Conformity depends on implementation. ICO provides the framework; individual deployers must conduct their own DPIA/PIA for specific use cases. See `docs/compliance/pia-summary.md` for the public PIA summary.

7. ISO Standards

Standard	Scope	DPP-CQ Reference
ISO 14067:2018	Carbon footprint of products	Sustainability module (sustainability.pcf)
ISO 22000:2018	Food safety management	Referenced in supply chain traceability
ISO 22739:2021	NFC and RFID standards	Carrier specification (L2/L3 NFC)
ISO 8601	Date and time format	All date fields in the schema
ISO 3166-1	Country codes	origin.country field

8. Multilateral Standards

Standard / Protocol	Scope	DPP-CQ Reference
W3C DID Core v1.1	Decentralized identifiers	did:ico:dpp method
W3C VC Data Model v2.0	Verifiable credentials	Core credential structure
IETF RFC 9529	SD-JWT VC	SD-JWT credential format profile
GS1 Digital Link 2.2	Supply chain identifiers	URI structure and resolver behavior
GS1 EPCIS 2.0	Event data for supply chain	Event mapping (§4 of gs1-digital-link-mapping.md)
UN/CEFACT UNTP	UN Transparency Protocol	Conformity claims vocabulary

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Annex K: PIA Summary

Public Document — ICO Std 2001 v2.0

Status: Public · Part of ICO Std 2001:2026 (DPP-CQ v2.0) **Maintained at:** docs/compliance/pia-summary.md
Version: 2.0.0-draft **Last updated:** September 2026

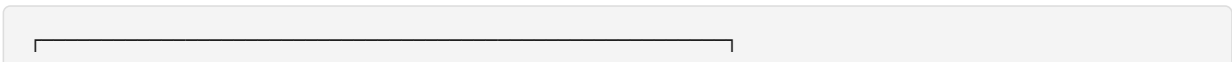
1. Purpose

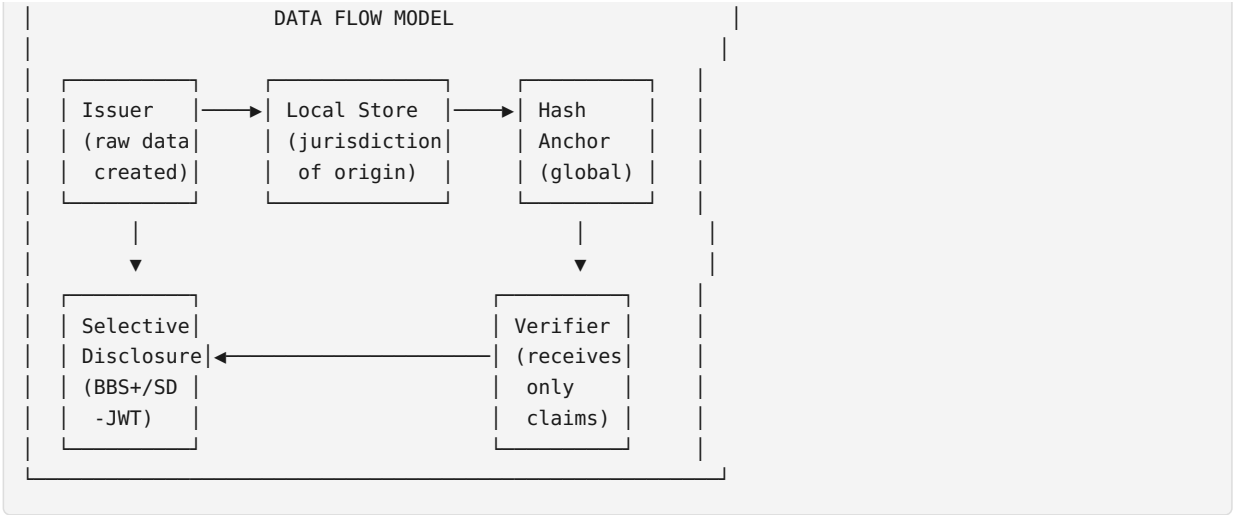
This document provides a public summary of the Privacy Impact Assessment (PIA) conducted for the DPP-CQ v2.0 standard framework. It is maintained as a normative reference per white paper §8.3.

This PIA summary addresses requirements under both the EU GDPR (Art. 35) and China’s PIPL (Art. 55).

2. Data Flow Architecture

DPP-CQ employs a “data localization, hash cross-border” architecture:





2.1 Data Categories

Category	Examples	Storage	Cross-Border
Product identity	Name, category, grade, GI number	Local + Hash anchor	Hash only
Origin data	Region, country, coordinates	Local	Hash only
Cultural data	Heritage status, craft method	Local	Hash only
Artisan personal data	Name, title, biography	Local (restricted)	Never
Supply chain data	Batch numbers, locations, entities	Local	Hash only
Quality attributes	Lab results, specifications	Local	Hash only
Sustainability data	PCF, material composition	Local	Hash only
Assessment data	AI disclosure, methodology refs	Local	Hash only
Lifecycle metadata	Controller, retention, privacy URI	Embedded in credential	Yes (public)

2.2 Cross-Border Data Transfer

Data Type	Transfer Method	Legal Basis
Hash fingerprints	Blockchain / distributed ledger	Hash = no identifiable information; not personal data
Credential metadata (public claims)	Verifiable Credential over HTTPS	Consent / legitimate interest
Selective disclosures	BBS+ / SD-JWT presentation	Consent (per disclosure)
Personal data (artisan info)	Never transferred cross-border	N/A (localized)

3. Risk Assessment

3.1 Risk Register

ID	Risk	Likelihood	Impact	Mitigation	Residual Risk
----	------	------------	--------	------------	---------------

R-01	Credential data leakage via resolver	Low	High	Hash-only cross-border; selective disclosure; encrypted storage	Low
R-02	Artisan personal data exposure	Medium	High	Restricted (R) disclosure class; localized storage; consent required	Low
R-03	Supply chain data competitive exposure	Medium	Medium	Holder-selectable disclosure; R-class for batch data	Low
R-04	QR code cloning (L1)	High	Low	Clone = same data; authenticity via cryptographic verification, not QR secrecy	Accepted
R-05	NFC tag spoofing (L2)	Low	High	SDM per-scan cryptographic authentication; AES-CMAC validation	Low
R-06	Cross-border data sovereignty violation	Low	Critical	"Data localization, hash cross-border" architecture; configurable jurisdiction	Low
R-07	Credential over-disclosure	Medium	Medium	Disclosure classification (P/H/R); holder controls presentation scope	Low
R-08	Retention policy violation	Low	Medium	<code>dataLifecycle</code> metadata; automated deletion triggers	Low

3.2 Risk Acceptance

All risks are assessed as **Low residual risk** after mitigation, except R-04 (QR cloning) which is an accepted risk by design — the security model does not rely on QR code secrecy.

4. Data Subject Rights

DPP-CQ supports the following data subject rights:

Right	GDPR Art.	PIPL Art.	DPP-CQ Implementation
Right to information	13-14	17	<code>dataLifecycle.privacyNoticeUri</code> in every credential
Right of access	15	45	Holder can retrieve full credential via DID resolution
Right to rectification	16	46	Issuer can issue updated credential; old version superseded

Right to erasure	17	47	Issuer can revoke credential; local data deletion per <code>dataLifecycle.deletionRules</code>
Right to restrict processing	18	—	Selective disclosure limits data exposure
Right to data portability	20	45	Standard VC format enables portability
Right to object	21	44	Holder controls which claims to disclose
Right to withdraw consent	7(3)	15	Revocation mechanism; artisan can withdraw consent for personal data disclosure

5. Cross-Border Legal Basis

5.1 EU GDPR

Transfer Mechanism	Applicability
Hash = non-personal data	Primary basis for hash cross-border
Standard Contractual Clauses (SCCs)	For any residual personal data transfers
Consent	For selective disclosure presentations
Legitimate interest	For public claims in verification flows

5.2 China PIPL

Transfer Mechanism	Applicability
Data localization	Raw data stored within China; only hashes cross borders
CAC security assessment	Not required (no personal data crosses border)
Standard contract	For any future cross-border personal data needs
Certification	ICO standard framework certification available

5.3 Other Jurisdictions

Jurisdiction	Approach
Southeast Asia	Flexible regional node deployment; configurable data storage
Middle East / GCC	Regional node deployment; no mandatory cross-border transfer
Global South	Lightweight solutions; data stays local

6. Data Lifecycle Metadata

Every DPP-CQ v2.0 credential carries a `dataLifecycle` record:

```
{
  "dataLifecycle": {
    "controller": "did:ico:issuer:zjtea-assoc",
```

```
"controllerContact": "privacy@zjtea-assoc.example",
"retentionPolicy": {
  "retentionPeriod": "P5Y",
  "basis": "contractual-obligation",
  "deletionTrigger": "expiration"
},
"storageJurisdiction": "CN",
"crossBorderMode": "hash-only",
"deletionRules": {
  "personalData": "deleted-within-30-days-of-withdrawal",
  "credentialData": "retained-for-verification-history",
  "hashAnchors": "permanent"
},
"privacyNoticeUri": "https://zjtea-assoc.example/privacy",
"piaSummaryUri": "https://resolver.icoun.org/pia/zjtea-assoc"
}
}
```

7. Governance

- This PIA summary is reviewed **annually** or upon significant standard changes
- The ICO Technical Committee maintains this document
- Stakeholder feedback is accepted via info@icoun.org
- Material changes require public review period (minimum 30 days)